

# Supplementary Information for

GenomeCF: a counterfactual validation standard for DNA sequence models

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# 1 Core and Screening Benchmark Tables

Table 1: GC-only diagnostic baselines on the focal human tasks.

| Task                | AUROC | ECE   | Brier |
|---------------------|-------|-------|-------|
| Enhancers<br>(Cohn) | 0.734 | 0.011 | 0.208 |
| Promoters           | 0.796 | 0.047 | 0.185 |

Table 2: Completed official-split real-data matrix for core and screening tasks.

|          | Tier | Task                | Model                           | Seeds | AUROC | ECE   | Brier | RC $\Delta$ | RC flip | Mono drop | Dinuc drop |
|----------|------|---------------------|---------------------------------|-------|-------|-------|-------|-------------|---------|-----------|------------|
| $\infty$ | Core | Promoters           | GC-only logistic regression     | 2     | 0.796 | 0.047 | 0.185 | 0.000       | 0.000   | 0.000     | 0.000      |
|          | Core | Promoters           | CpG-only logistic regression    | 1     | 0.844 | 0.048 | 0.157 | 0.000       | 0.000   | 0.000     | 0.000      |
|          | Core | Promoters           | Repeat-only logistic regression | 1     | 0.612 | 0.043 | 0.240 | 0.000       | 0.000   | 0.000     | 0.000      |
|          | Core | Promoters           | 6-mer logistic regression       | 2     | 0.898 | 0.110 | 0.133 | 0.263       | 0.268   | -0.003    | 0.058      |
|          | Core | Promoters           | CNN                             | 6     | 0.843 | 0.023 | 0.152 | 0.044       | 0.062   | -0.002    | 0.046      |
|          | Core | Promoters           | RC-aug CNN                      | 6     | 0.864 | 0.044 | 0.144 | 0.017       | 0.019   | -0.035    | 0.038      |
|          | Core | Promoters           | DNABERT-2                       | 2     | 0.888 | 0.042 | 0.128 | 0.078       | 0.073   | -0.054    | 0.004      |
|          | Core | Promoters           | Caduceus-Ph Nucleotide          | 1     | 0.853 | 0.024 | 0.154 | 0.040       | 0.035   | -0.060    | 0.002      |
|          | Core | Promoters           | Transformer v2                  | 1     | 0.649 | 0.112 | 0.249 | 0.270       | 0.455   | 0.015     | 0.018      |
|          | Core | Enhancers (Cohn)    | GC-only logistic regression     | 2     | 0.734 | 0.011 | 0.208 | 0.000       | 0.000   | 0.000     | 0.000      |
|          | Core | Enhancers (Cohn)    | CpG-only logistic regression    | 1     | 0.636 | 0.034 | 0.236 | 0.000       | 0.000   | 0.000     | 0.000      |
|          | Core | Enhancers (Cohn)    | Repeat-only logistic regression | 1     | 0.533 | 0.005 | 0.249 | 0.000       | 0.000   | 0.000     | 0.000      |
|          | Core | Enhancers (Cohn)    | 6-mer logistic regression       | 2     | 0.700 | 0.286 | 0.309 | 0.371       | 0.378   | -0.080    | 0.138      |
|          | Core | Enhancers (Cohn)    | CNN                             | 6     | 0.754 | 0.079 | 0.209 | 0.019       | 0.036   | 0.008     | 0.014      |
|          | Core | Enhancers (Cohn)    | RC-aug CNN                      | 6     | 0.765 | 0.071 | 0.203 | 0.014       | 0.031   | 0.003     | 0.017      |
|          | Core | Enhancers (Cohn)    | DNABERT-2                       | 2     | 0.830 | 0.025 | 0.168 | 0.059       | 0.097   | -0.049    | 0.023      |
|          | Core | Enhancers (Cohn)    | Caduceus-Ph Nucleotide          | 1     | 0.773 | 0.034 | 0.195 | 0.030       | 0.053   | -0.103    | -0.021     |
|          | Core | Enhancers (Cohn)    | Transformer v2                  | 1     | 0.518 | 0.164 | 0.284 | 0.228       | 0.507   | 0.017     | -0.013     |
|          | Core | Enhancers (Ensembl) | GC-only logistic regression     | 1     | 0.662 | 0.048 | 0.234 | 0.000       | 0.000   | 0.000     | 0.000      |
|          | Core | Enhancers (Ensembl) | CpG-only logistic regression    | 1     | 0.563 | 0.012 | 0.248 | 0.000       | 0.000   | 0.000     | 0.000      |
|          | Core | Enhancers (Ensembl) | Length-only logistic regression | 1     | 0.500 | 0.003 | 0.250 | 0.000       | 0.000   | 0.000     | 0.000      |
|          | Core | Enhancers (Ensembl) | Repeat-only logistic regression | 1     | 0.500 | 0.027 | 0.249 | 0.000       | 0.000   | 0.000     | 0.000      |
|          | Core | Enhancers (Ensembl) | 6-mer logistic regression       | 2     | 0.736 | 0.246 | 0.273 | 0.344       | 0.356   | 0.203     | 0.216      |
|          | Core | Enhancers (Ensembl) | CNN                             | 6     | 0.839 | 0.080 | 0.172 | 0.089       | 0.133   | 0.178     | 0.065      |

## 2 Split Robustness

Table 3: Five-fold chromosome-grouped cross-validation summary.

| Task                | Model                     | Mean AUROC | SD    | ECE   | Brier | RC $\Delta$ | Mono drop | Dinuc drop |
|---------------------|---------------------------|------------|-------|-------|-------|-------------|-----------|------------|
| Enhancers (Cohn)    | Caduceus-Ph               | 0.765      | 0.030 | 0.043 | 0.195 | 0.028       | -0.102    | -0.023     |
| Enhancers (Cohn)    | DNABERT-2                 | 0.827      | 0.021 | 0.028 | 0.168 | 0.065       | -0.007    | 0.023      |
| Enhancers (Cohn)    | 6-mer logistic regression | 0.724      | 0.017 | 0.217 | 0.263 | 0.305       | -0.066    | 0.095      |
| Enhancers (Cohn)    | CNN                       | 0.745      | 0.031 | 0.062 | 0.205 | 0.050       | 0.024     | 0.021      |
| Enhancers (Cohn)    | RC-aug CNN                | 0.771      | 0.030 | 0.057 | 0.195 | 0.057       | -0.005    | 0.040      |
| Enhancers (Ensembl) | Caduceus-Ph               | 0.717      | 0.008 | 0.046 | 0.217 | 0.045       | -0.040    | -0.011     |
| Enhancers (Ensembl) | 6-mer logistic regression | 0.770      | 0.006 | 0.182 | 0.232 | 0.283       | 0.192     | 0.229      |
| Enhancers (Ensembl) | CNN                       | 0.876      | 0.005 | 0.037 | 0.146 | 0.106       | 0.363     | 0.141      |
| Enhancers (Ensembl) | RC-aug CNN                | 0.886      | 0.001 | 0.038 | 0.139 | 0.087       | 0.386     | 0.168      |
| Promoters           | Caduceus-Ph               | 0.859      | 0.049 | 0.090 | 0.155 | 0.052       | -0.058    | 0.002      |
| Promoters           | DNABERT-2                 | 0.897      | 0.011 | 0.050 | 0.125 | 0.096       | -0.042    | 0.022      |
| Promoters           | 6-mer logistic regression | 0.834      | 0.003 | 0.226 | 0.240 | 0.269       | -0.005    | 0.041      |
| Promoters           | CNN                       | 0.869      | 0.021 | 0.063 | 0.144 | 0.093       | -0.022    | 0.053      |
| Promoters           | RC-aug CNN                | 0.873      | 0.018 | 0.096 | 0.159 | 0.070       | -0.061    | 0.038      |
| Open chromatin      | Caduceus-Ph               | 0.626      | 0.007 | 0.055 | 0.242 | 0.021       | -0.019    | -0.019     |
| Open chromatin      | 6-mer logistic regression | 0.682      | 0.002 | 0.215 | 0.278 | 0.314       | 0.212     | 0.114      |
| Open chromatin      | CNN                       | 0.695      | 0.014 | 0.061 | 0.225 | 0.093       | 0.067     | 0.017      |
| Open chromatin      | RC-aug CNN                | 0.734      | 0.004 | 0.043 | 0.211 | 0.115       | 0.139     | 0.049      |

Table 4: Per-fold chromosome-grouped cross-validation metrics and fold-wise counts.

|   | Task             | Model                     | Fold | AUROC | ECE   | Brier | RC $\Delta$ | Mono drop | Dinuc drop | P:N         | Test n |
|---|------------------|---------------------------|------|-------|-------|-------|-------------|-----------|------------|-------------|--------|
| 5 | Enhancers (Cohn) | Caduceus-Ph               | A    | 0.723 | 0.061 | 0.213 | 0.028       | -0.107    | -0.019     | 2,460:2,099 | 4,559  |
|   | Enhancers (Cohn) | Caduceus-Ph               | B    | 0.771 | 0.037 | 0.190 | 0.027       | -0.101    | -0.018     | 2,241:3,503 | 5,744  |
|   | Enhancers (Cohn) | Caduceus-Ph               | C    | 0.765 | 0.030 | 0.195 | 0.028       | -0.096    | -0.019     | 2,106:2,687 | 4,793  |
|   | Enhancers (Cohn) | Caduceus-Ph               | D    | 0.758 | 0.054 | 0.198 | 0.029       | -0.105    | -0.030     | 2,727:1,934 | 4,661  |
|   | Enhancers (Cohn) | Caduceus-Ph               | E    | 0.807 | 0.032 | 0.180 | 0.025       | -0.100    | -0.028     | 4,361:3,673 | 8,034  |
|   | Enhancers (Cohn) | DNABERT-2                 | A    | 0.797 | 0.051 | 0.184 | 0.066       | -0.033    | 0.027      | 2,460:2,099 | 4,559  |
|   | Enhancers (Cohn) | DNABERT-2                 | B    | 0.835 | 0.020 | 0.160 | 0.066       | -0.049    | 0.027      | 2,241:3,503 | 5,744  |
|   | Enhancers (Cohn) | DNABERT-2                 | C    | 0.827 | 0.024 | 0.169 | 0.065       | 0.026     | 0.023      | 2,106:2,687 | 4,793  |
|   | Enhancers (Cohn) | DNABERT-2                 | D    | 0.821 | 0.028 | 0.169 | 0.066       | -0.015    | 0.004      | 2,727:1,934 | 4,661  |
|   | Enhancers (Cohn) | DNABERT-2                 | E    | 0.853 | 0.017 | 0.156 | 0.062       | 0.038     | 0.033      | 4,361:3,673 | 8,034  |
|   | Enhancers (Cohn) | 6-mer logistic regression | A    | 0.706 | 0.242 | 0.283 | 0.306       | -0.091    | 0.099      | 2,460:2,099 | 4,559  |
|   | Enhancers (Cohn) | 6-mer logistic regression | B    | 0.715 | 0.216 | 0.262 | 0.304       | -0.085    | 0.091      | 2,241:3,503 | 5,744  |
|   | Enhancers (Cohn) | 6-mer logistic regression | C    | 0.727 | 0.219 | 0.264 | 0.304       | -0.066    | 0.106      | 2,106:2,687 | 4,793  |
|   | Enhancers (Cohn) | 6-mer logistic regression | D    | 0.722 | 0.213 | 0.261 | 0.311       | -0.009    | 0.104      | 2,727:1,934 | 4,661  |
|   | Enhancers (Cohn) | 6-mer logistic regression | E    | 0.752 | 0.196 | 0.245 | 0.303       | -0.079    | 0.073      | 4,361:3,673 | 8,034  |
|   | Enhancers (Cohn) | CNN                       | A    | 0.708 | 0.090 | 0.225 | 0.051       | 0.010     | 0.014      | 2,460:2,099 | 4,559  |
|   | Enhancers (Cohn) | CNN                       | B    | 0.734 | 0.077 | 0.206 | 0.051       | -0.002    | 0.014      | 2,241:3,503 | 5,744  |
|   | Enhancers (Cohn) | CNN                       | C    | 0.744 | 0.071 | 0.207 | 0.046       | 0.041     | 0.028      | 2,106:2,687 | 4,793  |
|   | Enhancers (Cohn) | CNN                       | D    | 0.745 | 0.032 | 0.201 | 0.056       | 0.036     | 0.027      | 2,727:1,934 | 4,661  |
|   | Enhancers (Cohn) | CNN                       | E    | 0.792 | 0.039 | 0.187 | 0.045       | 0.033     | 0.019      | 4,361:3,673 | 8,034  |
|   | Enhancers (Cohn) | RC-aug CNN                | A    | 0.733 | 0.067 | 0.212 | 0.059       | -0.007    | 0.040      | 2,460:2,099 | 4,559  |
|   | Enhancers (Cohn) | RC-aug CNN                | B    | 0.766 | 0.030 | 0.189 | 0.060       | -0.043    | 0.017      | 2,241:3,503 | 5,744  |
|   | Enhancers (Cohn) | RC-aug CNN                | C    | 0.774 | 0.021 | 0.181 | 0.059       | -0.000    | 0.064      | 2,106:2,687 | 4,793  |

### 3 Matched-negative and Mitigation Results

Table 5: Matched-negative evaluation on the focal real tasks.

| Task                | Model                       | Orig. AUROC | Matched AUROC | Orig. GC-only | Matched GC-only | Orig. ECE | Matched ECE | Orig. mono | Matched mono | Orig. dimuc | Matched dimuc |
|---------------------|-----------------------------|-------------|---------------|---------------|-----------------|-----------|-------------|------------|--------------|-------------|---------------|
| Promoters           | GC-only logistic regression | 0.796       | 0.712         | 0.796         | 0.712           | 0.047     | 0.122       | 0.000      | 0.000        | 0.000       | 0.000         |
| Promoters           | 6-mer logistic regression   | 0.916       | 0.911         | 0.796         | 0.712           | 0.095     | 0.113       | -0.013     | -0.008       | 0.049       | 0.058         |
| Promoters           | CNN                         | 0.891       | 0.873         | 0.796         | 0.712           | 0.024     | 0.128       | -0.034     | -0.059       | 0.049       | 0.042         |
| Promoters           | RC-aug CNN                  | 0.901       | 0.874         | 0.796         | 0.712           | 0.061     | 0.087       | -0.066     | -0.068       | 0.028       | 0.031         |
| Promoters           | DNABERT-2                   | 0.913       | 0.897         | 0.796         | 0.712           | 0.039     | 0.080       | -0.037     | -0.039       | 0.018       | 0.021         |
| Promoters           | Caduceus-Ph                 | 0.853       | 0.793         | 0.796         | 0.712           | 0.024     | 0.103       | -0.060     | -0.061       | 0.002       | 0.002         |
| Enhancers (Cohn)    | GC-only logistic regression | 0.734       | 0.639         | 0.734         | 0.639           | 0.011     | 0.069       | 0.000      | 0.000        | 0.000       | 0.000         |
| Enhancers (Cohn)    | 6-mer logistic regression   | 0.720       | 0.684         | 0.734         | 0.639           | 0.223     | 0.249       | -0.099     | -0.105       | 0.092       | 0.091         |
| Enhancers (Cohn)    | CNN                         | 0.750       | 0.673         | 0.734         | 0.639           | 0.037     | 0.074       | 0.025      | -0.006       | 0.005       | 0.015         |
| Enhancers (Cohn)    | RC-aug CNN                  | 0.756       | 0.696         | 0.734         | 0.639           | 0.035     | 0.081       | -0.020     | 0.016        | 0.004       | 0.013         |
| Enhancers (Cohn)    | DNABERT-2                   | 0.832       | 0.783         | 0.734         | 0.639           | 0.018     | 0.060       | -0.077     | -0.041       | 0.011       | 0.028         |
| Enhancers (Cohn)    | Caduceus-Ph                 | 0.773       | 0.693         | 0.734         | 0.639           | 0.034     | 0.073       | -0.103     | -0.102       | -0.021      | -0.021        |
| Enhancers (Ensembl) | GC-only logistic regression | 0.662       | 0.576         | 0.662         | 0.576           | 0.048     | 0.108       | 0.000      | 0.000        | 0.000       | 0.000         |
| Enhancers (Ensembl) | 6-mer logistic regression   | 0.778       | 0.759         | 0.662         | 0.576           | 0.175     | 0.196       | 0.202      | 0.209        | 0.256       | 0.253         |
| Enhancers (Ensembl) | CNN                         | 0.881       | 0.864         | 0.662         | 0.576           | 0.053     | 0.066       | 0.390      | 0.376        | 0.175       | 0.156         |
| Enhancers (Ensembl) | RC-aug CNN                  | 0.887       | 0.870         | 0.662         | 0.576           | 0.034     | 0.094       | 0.365      | 0.366        | 0.173       | 0.169         |
| Enhancers (Ensembl) | DNABERT-2                   | 0.775       | 0.774         | 0.662         | 0.576           | 0.027     | 0.076       | 0.043      | 0.092        | 0.009       | 0.034         |
| Enhancers (Ensembl) | Caduceus-Ph                 | 0.718       | 0.652         | 0.662         | 0.576           | 0.043     | 0.103       | -0.032     | -0.032       | -0.009      | -0.010        |

Table 6: Before-and-after confounder balance under matched-negative evaluation.

| Task                | GC neg/pos  |    | CpG neg/pos |    | Length neg/pos |    | Repeat neg/pos |    | N neg/pos   |    | GC-only AUROC |
|---------------------|-------------|----|-------------|----|----------------|----|----------------|----|-------------|----|---------------|
| Promoters           | 0.484/0.627 | -> | 0.265/0.682 | -> | 251.0/251.0    | -> | 0.222/0.243    | -> | 0.000/0.000 | -> | 0.796         |
|                     | 0.484/0.626 |    | 0.265/0.684 |    | 251.0/251.0    |    | 0.222/0.243    |    | 0.000/0.000 |    | 0.712         |
| Enhancers (Cohn)    | 0.394/0.469 | -> | 0.182/0.258 | -> | 500.0/500.0    | -> | 0.236/0.232    | -> | 0.000/0.000 | -> | 0.734         |
|                     | 0.394/0.469 |    | 0.182/0.258 |    | 500.0/500.0    |    | 0.236/0.232    |    | 0.000/0.000 |    | 0.639         |
| Enhancers (Ensembl) | 0.409/0.460 | -> | 0.209/0.249 | -> | 267.9/267.9    | -> | 0.233/0.245    | -> | 0.001/0.000 | -> | 0.662         |
|                     | 0.409/0.460 |    | 0.209/0.249 |    | 267.9/267.9    |    | 0.233/0.245    |    | 0.001/0.000 |    | 0.576         |

Table 7: Completed mitigation experiments, including matched-negative retraining and completed frozen-foundation interventions.

| Task                | Model       | Intervention                | AUROC before | AUROC after | ECE before | ECE after | Brier before | Brier after | RC before | RC after | Mono before | Mono after | Dimuc before | Dimuc after | Matched AUROC before | Matched AUROC after | Matched ECE before | Matched ECE after |
|---------------------|-------------|-----------------------------|--------------|-------------|------------|-----------|--------------|-------------|-----------|----------|-------------|------------|--------------|-------------|----------------------|---------------------|--------------------|-------------------|
| Promoters           | CNN         | Temperature scaling         | 0.891        | 0.875       | 0.024      | 0.039     | 0.129        | 0.135       | 0.081     | 0.065    | -0.034      | 0.000      | 0.049        | 0.044       | 0.873                | –                   | 0.128              | –                 |
| Promoters           | CNN         | GC-balanced training        | 0.891        | 0.874       | 0.024      | 0.132     | 0.129        | 0.163       | 0.081     | 0.064    | -0.034      | 0.028      | 0.049        | 0.058       | 0.873                | –                   | 0.128              | –                 |
| Promoters           | CNN         | Matched-negative retraining | 0.891        | 0.871       | 0.024      | 0.067     | 0.129        | 0.142       | 0.081     | 0.092    | -0.034      | -0.037     | 0.049        | 0.052       | 0.873                | 0.854               | 0.128              | 0.012             |
| Promoters           | RC-aug CNN  | Temperature scaling         | 0.901        | 0.897       | 0.061      | 0.030     | 0.124        | 0.119       | 0.050     | 0.063    | -0.066      | -0.033     | 0.028        | 0.042       | 0.874                | –                   | 0.087              | –                 |
| Promoters           | RC-aug CNN  | GC-balanced training        | 0.901        | 0.901       | 0.061      | 0.040     | 0.124        | 0.117       | 0.050     | 0.056    | -0.066      | -0.052     | 0.028        | 0.039       | 0.874                | –                   | 0.087              | –                 |
| Promoters           | RC-aug CNN  | Matched-negative retraining | 0.901        | 0.882       | 0.061      | 0.078     | 0.124        | 0.139       | 0.050     | 0.046    | -0.066      | -0.062     | 0.028        | 0.045       | 0.874                | 0.893               | 0.087              | 0.049             |
| Promoters           | DNABERT-2   | Temperature scaling         | 0.913        | 0.913       | 0.039      | 0.020     | 0.111        | 0.113       | 0.082     | 0.090    | -0.037      | -0.049     | 0.018        | 0.021       | 0.897                | 0.897               | 0.080              | 0.081             |
| Promoters           | DNABERT-2   | Matched-negative retraining | 0.913        | 0.908       | 0.039      | 0.112     | 0.111        | 0.137       | 0.082     | 0.098    | -0.037      | -0.049     | 0.018        | 0.013       | 0.897                | 0.910               | 0.080              | 0.027             |
| Promoters           | Caduceus-Ph | Temperature scaling         | 0.853        | 0.853       | 0.024      | 0.031     | 0.154        | 0.155       | 0.040     | 0.048    | -0.060      | -0.068     | 0.002        | 0.002       | 0.793                | 0.793               | 0.103              | 0.098             |
| Promoters           | Caduceus-Ph | Matched-negative retraining | 0.853        | 0.862       | 0.024      | 0.140     | 0.154        | 0.175       | 0.040     | 0.048    | -0.060      | -0.049     | 0.002        | -0.000      | 0.793                | 0.811               | 0.103              | 0.046             |
| Enhancers (Cohn)    | CNN         | Temperature scaling         | 0.750        | 0.761       | 0.037      | 0.040     | 0.204        | 0.200       | 0.046     | 0.028    | 0.025       | 0.020      | 0.005        | 0.004       | 0.673                | –                   | 0.074              | –                 |
| Enhancers (Cohn)    | CNN         | GC-balanced training        | 0.750        | 0.752       | 0.037      | 0.120     | 0.204        | 0.219       | 0.046     | 0.033    | 0.025       | 0.001      | 0.005        | -0.002      | 0.673                | –                   | 0.074              | –                 |
| Enhancers (Cohn)    | CNN         | Matched-negative retraining | 0.750        | 0.739       | 0.037      | 0.109     | 0.204        | 0.228       | 0.046     | 0.028    | 0.025       | -0.013     | 0.005        | 0.020       | 0.673                | 0.672               | 0.074              | 0.059             |
| Enhancers (Cohn)    | RC-aug CNN  | Temperature scaling         | 0.756        | 0.765       | 0.035      | 0.020     | 0.202        | 0.198       | 0.036     | 0.065    | -0.020      | 0.020      | 0.004        | 0.022       | 0.696                | –                   | 0.081              | –                 |
| Enhancers (Cohn)    | RC-aug CNN  | GC-balanced training        | 0.756        | 0.767       | 0.035      | 0.062     | 0.202        | 0.201       | 0.036     | 0.041    | -0.020      | 0.013      | 0.004        | 0.012       | 0.696                | –                   | 0.081              | –                 |
| Enhancers (Cohn)    | RC-aug CNN  | Matched-negative retraining | 0.756        | 0.757       | 0.035      | 0.102     | 0.202        | 0.218       | 0.036     | 0.044    | -0.020      | -0.008     | 0.004        | 0.016       | 0.696                | 0.683               | 0.081              | 0.019             |
| Enhancers (Cohn)    | DNABERT-2   | Temperature scaling         | 0.832        | 0.833       | 0.018      | 0.019     | 0.167        | 0.167       | 0.060     | 0.065    | -0.077      | -0.042     | 0.011        | 0.029       | 0.783                | 0.783               | 0.060              | 0.061             |
| Enhancers (Cohn)    | DNABERT-2   | Matched-negative retraining | 0.832        | 0.820       | 0.018      | 0.073     | 0.167        | 0.180       | 0.060     | 0.070    | -0.077      | -0.110     | 0.011        | 0.014       | 0.783                | 0.789               | 0.060              | 0.016             |
| Enhancers (Cohn)    | Caduceus-Ph | Temperature scaling         | 0.773        | 0.774       | 0.034      | 0.037     | 0.195        | 0.194       | 0.030     | 0.031    | -0.103      | -0.112     | -0.021       | -0.024      | 0.693                | 0.696               | 0.073              | 0.076             |
| Enhancers (Cohn)    | Caduceus-Ph | Matched-negative retraining | 0.773        | 0.777       | 0.034      | 0.089     | 0.195        | 0.206       | 0.030     | 0.037    | -0.103      | -0.105     | -0.021       | -0.024      | 0.693                | 0.711               | 0.073              | 0.029             |
| Enhancers (Ensembl) | CNN         | GC-balanced training        | 0.881        | 0.873       | 0.053      | 0.092     | 0.144        | 0.157       | 0.101     | 0.119    | 0.390       | 0.375      | 0.175        | 0.178       | 0.864                | –                   | 0.066              | –                 |
| Enhancers (Ensembl) | CNN         | Matched-negative retraining | 0.881        | 0.859       | 0.053      | 0.148     | 0.144        | 0.179       | 0.101     | 0.109    | 0.390       | 0.254      | 0.175        | 0.115       | 0.864                | 0.849               | 0.066              | 0.072             |
| Enhancers (Ensembl) | RC-aug CNN  | GC-balanced training        | 0.887        | 0.886       | 0.034      | 0.047     | 0.138        | 0.140       | 0.088     | 0.092    | 0.365       | 0.358      | 0.173        | 0.166       | 0.870                | –                   | 0.094              | –                 |
| Enhancers (Ensembl) | RC-aug CNN  | Matched-negative retraining | 0.887        | 0.874       | 0.034      | 0.098     | 0.138        | 0.157       | 0.088     | 0.077    | 0.365       | 0.336      | 0.173        | 0.121       | 0.870                | 0.864               | 0.094              | 0.054             |

## 4 GC-bin Robustness

Table 8: GC-bin robustness summary on the core human tasks.

| Task                | Model                     | Overall AUROC | Worst-bin AUROC | AUROC gap | Overall ECE | Worst-bin ECE | ECE gap |
|---------------------|---------------------------|---------------|-----------------|-----------|-------------|---------------|---------|
| Promoters           | 6-mer logistic regression | 0.916         | 0.767           | 0.218     | 0.095       | 0.142         | 0.125   |
| Promoters           | CNN                       | 0.901         | 0.664           | 0.307     | 0.087       | 0.139         | 0.130   |
| Promoters           | RC-aug CNN                | 0.902         | 0.652           | 0.335     | 0.060       | 0.106         | 0.080   |
| Promoters           | DNABERT-2                 | 0.913         | 0.718           | 0.262     | 0.020       | 0.038         | 0.010   |
| Promoters           | Caduceus-Ph               | 0.853         | 0.664           | 0.208     | 0.024       | 0.104         | 0.065   |
| Enhancers (Cohn)    | 6-mer logistic regression | 0.720         | 0.631           | 0.047     | 0.223       | 0.275         | 0.085   |
| Enhancers (Cohn)    | CNN                       | 0.745         | 0.592           | 0.068     | 0.098       | 0.132         | 0.086   |
| Enhancers (Cohn)    | RC-aug CNN                | 0.772         | 0.667           | 0.023     | 0.057       | 0.087         | 0.053   |
| Enhancers (Cohn)    | DNABERT-2                 | 0.833         | 0.752           | 0.055     | 0.016       | 0.039         | 0.021   |
| Enhancers (Cohn)    | Caduceus-Ph               | 0.774         | 0.656           | 0.082     | 0.036       | 0.074         | 0.046   |
| Enhancers (Ensembl) | 6-mer logistic regression | 0.778         | 0.731           | 0.053     | 0.175       | 0.193         | 0.045   |
| Enhancers (Ensembl) | CNN                       | 0.878         | 0.835           | 0.072     | 0.022       | 0.048         | 0.026   |
| Enhancers (Ensembl) | RC-aug CNN                | 0.888         | 0.846           | 0.062     | 0.012       | 0.041         | 0.023   |
| Enhancers (Ensembl) | DNABERT-2                 | 0.807         | 0.761           | 0.027     | 0.021       | 0.050         | 0.020   |
| Enhancers (Ensembl) | Caduceus-Ph               | 0.720         | 0.617           | 0.064     | 0.044       | 0.074         | 0.031   |
| Open chromatin      | 6-mer logistic regression | 0.686         | 0.670           | 0.015     | 0.219       | 0.242         | 0.053   |
| Open chromatin      | CNN                       | 0.697         | 0.663           | 0.034     | 0.042       | 0.080         | 0.062   |
| Open chromatin      | RC-aug CNN                | 0.737         | 0.707           | 0.052     | 0.029       | 0.071         | 0.056   |
| Open chromatin      | DNABERT-2                 | 0.718         | 0.690           | 0.020     | 0.011       | 0.048         | 0.024   |
| Open chromatin      | Caduceus-Ph               | 0.625         | 0.558           | 0.076     | 0.057       | 0.096         | 0.061   |

Table 9: Full GC-bin metrics by task, model and composition bin.

|   | Task             | Model                     | GC bin | GC mean | n     | P:N       | AUROC | ECE   | Brier |
|---|------------------|---------------------------|--------|---------|-------|-----------|-------|-------|-------|
| c | Promoters        | 6-mer logistic regression | 0      | 0.361   | 1,843 | 469:1,374 | 0.767 | 0.142 | 0.169 |
|   | Promoters        | 6-mer logistic regression | 1      | 0.479   | 1,801 | 557:1,244 | 0.820 | 0.135 | 0.164 |
|   | Promoters        | 6-mer logistic regression | 2      | 0.568   | 1,811 | 941:870   | 0.878 | 0.130 | 0.157 |
|   | Promoters        | 6-mer logistic regression | 3      | 0.653   | 1,792 | 1,282:510 | 0.940 | 0.072 | 0.091 |
|   | Promoters        | 6-mer logistic regression | 4      | 0.755   | 1,787 | 1,666:121 | 0.985 | 0.017 | 0.023 |
|   | Promoters        | CNN                       | 0      | 0.361   | 1,843 | 469:1,374 | 0.664 | 0.120 | 0.190 |
|   | Promoters        | CNN                       | 1      | 0.479   | 1,801 | 557:1,244 | 0.785 | 0.120 | 0.181 |
|   | Promoters        | CNN                       | 2      | 0.568   | 1,811 | 941:870   | 0.878 | 0.139 | 0.166 |
|   | Promoters        | CNN                       | 3      | 0.653   | 1,792 | 1,282:510 | 0.933 | 0.064 | 0.101 |
|   | Promoters        | CNN                       | 4      | 0.755   | 1,787 | 1,666:121 | 0.971 | 0.009 | 0.027 |
|   | Promoters        | RC-aug CNN                | 0      | 0.361   | 1,843 | 469:1,374 | 0.652 | 0.055 | 0.179 |
|   | Promoters        | RC-aug CNN                | 1      | 0.479   | 1,801 | 557:1,244 | 0.790 | 0.070 | 0.164 |
|   | Promoters        | RC-aug CNN                | 2      | 0.568   | 1,811 | 941:870   | 0.877 | 0.106 | 0.153 |
|   | Promoters        | RC-aug CNN                | 3      | 0.653   | 1,792 | 1,282:510 | 0.934 | 0.064 | 0.098 |
|   | Promoters        | RC-aug CNN                | 4      | 0.755   | 1,787 | 1,666:121 | 0.987 | 0.026 | 0.022 |
|   | Promoters        | DNABERT-2                 | 0      | 0.361   | 1,843 | 469:1,374 | 0.718 | 0.036 | 0.167 |
|   | Promoters        | DNABERT-2                 | 1      | 0.479   | 1,801 | 557:1,244 | 0.817 | 0.031 | 0.149 |
|   | Promoters        | DNABERT-2                 | 2      | 0.568   | 1,811 | 941:870   | 0.891 | 0.038 | 0.133 |
|   | Promoters        | DNABERT-2                 | 3      | 0.653   | 1,792 | 1,282:510 | 0.935 | 0.033 | 0.092 |
|   | Promoters        | DNABERT-2                 | 4      | 0.755   | 1,787 | 1,666:121 | 0.980 | 0.028 | 0.024 |
|   | Promoters        | Caduceus-Ph               | 0      | 0.361   | 1,843 | 469:1,374 | 0.664 | 0.066 | 0.185 |
|   | Promoters        | Caduceus-Ph               | 1      | 0.479   | 1,801 | 557:1,244 | 0.740 | 0.074 | 0.186 |
|   | Promoters        | Caduceus-Ph               | 2      | 0.568   | 1,811 | 941:870   | 0.819 | 0.104 | 0.186 |
|   | Promoters        | Caduceus-Ph               | 3      | 0.653   | 1,792 | 1,282:510 | 0.829 | 0.095 | 0.156 |
|   | Promoters        | Caduceus-Ph               | 4      | 0.755   | 1,787 | 1,666:121 | 0.873 | 0.038 | 0.054 |
|   | Enhancers (Cohn) | 6-mer logistic regression | 0      | 0.319   | 1,427 | 328:1,099 | 0.664 | 0.206 | 0.231 |
|   | Enhancers (Cohn) | 6-mer logistic regression | 1      | 0.378   | 1,372 | 523:849   | 0.662 | 0.240 | 0.287 |

## 5 External Biological Validation

Table 10: Full external biological validation matrix.

| External task            | Model                     | Official AUROC | AUPRC | ECE   | Brier | Matched AUROC | Matched shift | Worst-bin AUROC | GC-bin gap |
|--------------------------|---------------------------|----------------|-------|-------|-------|---------------|---------------|-----------------|------------|
| External TF (human_tf_0) | 6-mer logistic regression | 0.782          | 0.742 | 0.151 | 0.213 | 0.804         | -0.022        | 0.742           | 0.117      |
| External TF (human_tf_0) | CNN                       | 0.728          | 0.689 | 0.143 | 0.229 | 0.873         | -0.145        | 0.680           | 0.099      |
| External TF (human_tf_0) | RC-aug CNN                | 0.810          | 0.762 | 0.184 | 0.226 | 0.873         | -0.063        | 0.753           | 0.159      |
| External TF (human_tf_0) | DNABERT-2                 | 0.824          | 0.817 | 0.093 | 0.179 | 0.880         | -0.056        | 0.761           | 0.175      |
| External TF (human_tf_0) | DNABERT-2                 | 0.846          | 0.844 | 0.061 | 0.165 | 0.883         | -0.038        | 0.761           | 0.175      |
| External TF (human_tf_0) | DNABERT-2                 | 0.846          | 0.844 | 0.067 | 0.164 | 0.883         | -0.038        | 0.761           | 0.175      |
| External TF (human_tf_0) | Caduceus-Ph               | 0.716          | 0.674 | 0.121 | 0.239 | 0.810         | -0.094        | 0.721           | 0.150      |
| External TF (human_tf_0) | Caduceus-Ph               | 0.758          | 0.705 | 0.155 | 0.232 | 0.823         | -0.064        | 0.721           | 0.150      |
| External TF (human_tf_0) | Caduceus-Ph               | 0.758          | 0.705 | 0.130 | 0.222 | 0.784         | -0.026        | 0.721           | 0.150      |
| External TF (human_tf_1) | 6-mer logistic regression | 0.863          | 0.832 | 0.097 | 0.157 | 0.849         | 0.014         | 0.831           | 0.071      |
| External TF (human_tf_1) | CNN                       | 0.794          | 0.783 | 0.136 | 0.205 | 0.885         | -0.091        | 0.711           | 0.102      |
| External TF (human_tf_1) | RC-aug CNN                | 0.886          | 0.875 | 0.132 | 0.155 | 0.899         | -0.012        | 0.868           | 0.076      |
| External TF (human_tf_1) | DNABERT-2                 | 0.923          | 0.906 | 0.119 | 0.126 | 0.909         | 0.014         | 0.877           | 0.070      |
| External TF (human_tf_1) | DNABERT-2                 | 0.924          | 0.921 | 0.079 | 0.120 | -             | -             | 0.877           | 0.070      |
| External TF (human_tf_1) | Caduceus-Ph               | 0.802          | 0.784 | 0.188 | 0.220 | 0.853         | -0.051        | 0.817           | 0.105      |
| External H3K4me3         | 6-mer logistic regression | 0.586          | 0.598 | 0.299 | 0.348 | 0.622         | -0.036        | 0.536           | 0.103      |
| External H3K4me3         | CNN                       | 0.594          | 0.574 | 0.021 | 0.249 | 0.589         | 0.006         | 0.489           | 0.200      |
| External H3K4me3         | RC-aug CNN                | 0.595          | 0.595 | 0.015 | 0.250 | 0.607         | -0.012        | 0.510           | 0.182      |
| External H3K4me3         | DNABERT-2                 | 0.648          | 0.643 | 0.046 | 0.234 | 0.663         | -0.015        | 0.569           | 0.121      |
| External H3K4me3         | DNABERT-2                 | 0.666          | 0.682 | 0.028 | 0.229 | -             | -             | 0.569           | 0.121      |
| External H3K4me3         | Caduceus-Ph               | 0.577          | 0.556 | 0.028 | 0.246 | 0.570         | 0.008         | 0.478           | 0.175      |
| External H3K14ac         | 6-mer logistic regression | 0.665          | 0.678 | 0.237 | 0.294 | 0.686         | -0.021        | 0.571           | 0.145      |
| External H3K14ac         | CNN                       | 0.621          | 0.619 | 0.033 | 0.249 | 0.694         | -0.073        | 0.623           | 0.122      |
| External H3K14ac         | RC-aug CNN                | 0.634          | 0.638 | 0.023 | 0.239 | 0.713         | -0.079        | 0.624           | 0.098      |
| External H3K14ac         | DNABERT-2                 | 0.755          | 0.756 | 0.037 | 0.202 | 0.761         | -0.006        | 0.617           | 0.213      |
| External H3K14ac         | DNABERT-2                 | 0.762          | 0.800 | 0.064 | 0.199 | -             | -             | 0.617           | 0.213      |
| External H3K14ac         | Caduceus-Ph               | 0.591          | 0.613 | 0.045 | 0.245 | 0.597         | -0.006        | 0.430           | 0.157      |
| MPRA BCL11A enhancer     | 6-mer logistic regression | 0.463          | 0.080 | 0.061 | 0.093 | -             | -             | 0.331           | 0.195      |
| MPRA BCL11A enhancer     | CNN                       | 0.434          | 0.097 | 0.039 | 0.074 | -             | -             | 0.369           | 0.223      |
| MPRA BCL11A enhancer     | CNN                       | 0.583          | 0.167 | 0.042 | 0.074 | -             | -             | 0.539           | 0.156      |
| MPRA BCL11A enhancer     | RC-aug CNN                | 0.441          | 0.073 | 0.036 | 0.074 | -             | -             | 0.431           | 0.071      |
| MPRA BCL11A enhancer     | RC-aug CNN                | 0.441          | 0.073 | 0.041 | 0.074 | -             | -             | 0.431           | 0.071      |
| MPRA BCL11A enhancer     | DNABERT-2                 | 0.507          | 0.083 | 0.423 | 0.251 | -             | -             | 0.489           | 0.037      |
| MPRA BCL11A enhancer     | DNABERT-2                 | 0.540          | 0.125 | 0.052 | 0.075 | -             | -             | 0.340           | 0.275      |
| MPRA BCL11A enhancer     | DNABERT-2                 | 0.540          | 0.125 | 0.041 | 0.074 | -             | -             | 0.340           | 0.275      |
| MPRA BCL11A enhancer     | Caduceus-Ph               | 0.583          | 0.090 | 0.422 | 0.250 | -             | -             | 0.328           | 0.393      |
| MPRA BCL11A enhancer     | Caduceus-Ph               | 0.431          | 0.074 | 0.051 | 0.075 | -             | -             | 0.323           | 0.154      |
| MPRA BCL11A enhancer     | Caduceus-Ph               | 0.431          | 0.074 | 0.042 | 0.074 | -             | -             | 0.323           | 0.154      |
| MPRA F9 promoter         | 6-mer logistic regression | 0.672          | 0.726 | 0.153 | 0.234 | -             | -             | 0.614           | 0.096      |
| MPRA F9 promoter         | CNN                       | 0.524          | 0.520 | 0.021 | 0.250 | -             | -             | 0.448           | 0.135      |
| MPRA F9 promoter         | CNN                       | 0.524          | 0.520 | 0.034 | 0.251 | -             | -             | 0.448           | 0.135      |
| MPRA F9 promoter         | RC-aug CNN                | 0.468          | 0.459 | 0.003 | 0.250 | -             | -             | 0.445           | 0.014      |
| MPRA F9 promoter         | RC-aug CNN                | 0.468          | 0.459 | 0.002 | 0.250 | -             | -             | 0.445           | 0.014      |
| MPRA F9 promoter         | DNABERT-2                 | 0.543          | 0.547 | 0.051 | 0.249 | -             | -             | 0.477           | 0.154      |
| MPRA F9 promoter         | DNABERT-2                 | 0.524          | 0.534 | 0.018 | 0.249 | -             | -             | 0.451           | 0.179      |
| MPRA F9 promoter         | DNABERT-2                 | 0.524          | 0.534 | 0.016 | 0.249 | -             | -             | 0.451           | 0.179      |
| MPRA F9 promoter         | Caduceus-Ph               | 0.546          | 0.550 | 0.037 | 0.250 | -             | -             | 0.521           | 0.014      |
| MPRA F9 promoter         | Caduceus-Ph               | 0.579          | 0.535 | 0.003 | 0.249 | -             | -             | 0.533           | 0.040      |
| MPRA F9 promoter         | Caduceus-Ph               | 0.579          | 0.535 | 0.004 | 0.250 | -             | -             | 0.533           | 0.040      |
| MPRA HBB promoter        | 6-mer logistic regression | 0.714          | 0.624 | 0.051 | 0.210 | -             | -             | 0.624           | 0.127      |
| MPRA HBB promoter        | CNN                       | 0.580          | 0.540 | 0.057 | 0.242 | -             | -             | 0.248           | 0.434      |
| MPRA HBB promoter        | CNN                       | 0.580          | 0.540 | 0.077 | 0.244 | -             | -             | 0.248           | 0.434      |
| MPRA HBB promoter        | RC-aug CNN                | 0.622          | 0.535 | 0.042 | 0.240 | -             | -             | 0.547           | 0.090      |
| MPRA HBB promoter        | RC-aug CNN                | 0.622          | 0.535 | 0.078 | 0.245 | -             | -             | 0.547           | 0.090      |
| MPRA HBB promoter        | DNABERT-2                 | 0.488          | 0.385 | 0.099 | 0.251 | -             | -             | 0.444           | 0.059      |
| MPRA HBB promoter        | DNABERT-2                 | 0.485          | 0.378 | 0.059 | 0.243 | -             | -             | 0.444           | 0.041      |
| MPRA HBB promoter        | DNABERT-2                 | 0.485          | 0.378 | 0.060 | 0.244 | -             | -             | 0.444           | 0.041      |

Table 11: External transfer prediction statistics used in the main-text correlation analysis.

| Analysis                                                   | Statistic                  | Estimate (95% CI)      | n  |
|------------------------------------------------------------|----------------------------|------------------------|----|
| Core AUROC vs external biological reliability              | Spearman                   | 0.215 [-0.018, 0.428]  | 82 |
| GenomeCF Shortcut Score vs external biological reliability | Spearman                   | 0.077 [-0.129, 0.271]  | 82 |
| Core AUROC vs external reliability risk                    | Spearman                   | -0.043 [-0.278, 0.171] | 82 |
| GenomeCF Shortcut Score vs external reliability risk       | Spearman                   | -0.166 [-0.384, 0.053] | 82 |
| Core AUROC vs external matched-negative shift              | Pearson                    | -0.142 [-0.460, 0.152] | 24 |
| GenomeCF Shortcut Score vs external matched-negative shift | Pearson                    | 0.283 [-0.121, 0.615]  | 24 |
| Held-out core AUROC model                                  | $R^2$                      | 0.062 [0.001, 0.215]   | 82 |
| GenomeCF Shortcut Score model                              | $R^2$                      | 0.023 [0.000, 0.111]   | 82 |
| Combined AUROC + Shortcut Score model                      | $R^2$                      | 0.086 [0.020, 0.239]   | 82 |
| Full GenomeCF profile model                                | $R^2$                      | 0.383 [0.263, 0.564]   | 82 |
| Histone marks family: Held-out core AUROC model            | Family-stratified $R^2$    | 0.614 [0.361, 0.855]   | 12 |
| TF binding family: Held-out core AUROC model               | Family-stratified $R^2$    | 0.389 [0.033, 0.753]   | 15 |
| Variant effect family: Held-out core AUROC model           | Family-stratified $R^2$    | 0.000 [0.000, 0.097]   | 55 |
| Histone marks family: Full GenomeCF profile model          | Family-stratified $R^2$    | 0.958 [0.913, 1.000]   | 12 |
| TF binding family: Full GenomeCF profile model             | Family-stratified $R^2$    | 0.845 [0.743, 0.999]   | 15 |
| Variant effect family: Full GenomeCF profile model         | Family-stratified $R^2$    | 0.356 [0.185, 0.617]   | 55 |
| Held-out core AUROC model                                  | Leave-one-family-out $R^2$ | -0.318                 | 82 |
| GenomeCF Shortcut Score model                              | Leave-one-family-out $R^2$ | -0.037                 | 82 |
| Combined AUROC + Shortcut Score model                      | Leave-one-family-out $R^2$ | -0.321                 | 82 |
| Full GenomeCF profile model                                | Leave-one-family-out $R^2$ | -0.109                 | 82 |

Table 12: External prediction robustness analysis across task- and family-stratified settings.

| Analysis                                             | n  | AUROC-only R <sup>2</sup> | Shortcut-only R <sup>2</sup> | Full profile R <sup>2</sup> | $\Delta R^2$ (95% CI) | Perm. <i>p</i> | Notes                                                           |
|------------------------------------------------------|----|---------------------------|------------------------------|-----------------------------|-----------------------|----------------|-----------------------------------------------------------------|
| in-sample (pooled)                                   | 82 | 0.062                     | 0.023                        | 0.383                       | 0.321 [0.182, 0.515]  | 0.0005         | Primary analysis: full GenomeCF profile vs AUROC-only (pooled). |
| leave-one-task-out                                   | 82 | 0.026                     | -0.027                       | 0.238                       | 0.212                 | –              | Task-held-out prediction within the same mix of assay families. |
| leave-one-family-out                                 | 82 | -0.318                    | -0.037                       | -0.109                      | 0.208                 | 0.3616         | Hard setting: extrapolation to a held-out assay family.         |
| within-family leave-one-task-out (Variant effect)    | 55 | -0.025                    | -0.060                       | 0.089                       | 0.114                 | –              | Within-family robustness (assay-family specific).               |
| in-sample model-family stratified (k-mer baseline)   | 9  | 0.000                     | 0.000                        | 0.000                       | 0.000                 | –              | Checks robustness when restricting to a model family.           |
| in-sample model-family stratified (CNN)              | 28 | 0.006                     | 0.028                        | 0.076                       | 0.069                 | –              | Checks robustness when restricting to a model family.           |
| in-sample model-family stratified (foundation model) | 45 | 0.247                     | 0.086                        | 0.555                       | 0.308                 | –              | Checks robustness when restricting to a model family.           |

Table 13: External biological case-study results for the BCL11A and MYC MPRA regulatory variant-effect tasks.

| Case   | Task                 | Model                     | Config.            | Decision role                         | Core AUROC | Core ECE | Core score | External AUROC | External AUPRC | Spearman[ $\Delta$ ] | Top-k enrich. | ECE   | Brier | Worst-bin | GC gap |
|--------|----------------------|---------------------------|--------------------|---------------------------------------|------------|----------|------------|----------------|----------------|----------------------|---------------|-------|-------|-----------|--------|
| Case A | MPRA BCL11A enhancer | CNN                       | Standard           | AUROC-only default configuration      | 0.769      | 0.061    | 0.451      | 0.434          | 0.097          | -0.078               | 1.234         | 0.039 | 0.074 | 0.369     | 0.223  |
| Case A | MPRA BCL11A enhancer | CNN                       | Temperature-scaled | GenomeCF-aware configuration          | 0.818      | 0.040    | 0.443      | 0.583          | 0.167          | 0.013                | 2.056         | 0.042 | 0.074 | 0.539     | 0.156  |
| Case A | MPRA BCL11A enhancer | DNABERT-2                 | Standard           | Reference comparison                  | 0.794      | 0.029    | 0.637      | 0.540          | 0.125          | 0.007                | 1.234         | 0.052 | 0.075 | 0.340     | 0.275  |
| Case B | MPRA MYC enhancer    | DNABERT-2                 | Standard           | GenomeCF-aware variant-ranking choice | 0.794      | 0.029    | 0.637      | 0.580          | 0.221          | 0.105                | 1.934         | 0.002 | 0.131 | 0.503     | 0.207  |
| Case B | MPRA MYC enhancer    | 6-mer logistic regression | Standard           | AUROC-only variant-ranking choice     | 0.746      | 0.235    | 0.646      | 0.617          | 0.273          | 0.130                | 1.504         | 0.070 | 0.133 | 0.579     | 0.103  |

Table 14: External GC-bin robustness summary.

| External task            | Model                     | Overall AUROC | Worst-bin AUROC | AUROC gap | Overall ECE | Worst-bin ECE | ECE gap |
|--------------------------|---------------------------|---------------|-----------------|-----------|-------------|---------------|---------|
| External H3K14ac         | Caduceus-Ph               | 0.591         | 0.430           | 0.157     | 0.045       | 0.125         | 0.118   |
| External H3K14ac         | DNABERT-2                 | 0.755         | 0.617           | 0.213     | 0.037       | 0.137         | 0.061   |
| External H3K14ac         | 6-mer logistic regression | 0.665         | 0.571           | 0.145     | 0.237       | 0.349         | 0.170   |
| External H3K14ac         | CNN                       | 0.680         | 0.623           | 0.122     | 0.030       | 0.140         | 0.121   |
| External H3K14ac         | RC-aug CNN                | 0.643         | 0.624           | 0.098     | 0.066       | 0.158         | 0.075   |
| External H3K4me3         | Caduceus-Ph               | 0.577         | 0.478           | 0.175     | 0.028       | 0.047         | 0.040   |
| External H3K4me3         | DNABERT-2                 | 0.648         | 0.569           | 0.121     | 0.046       | 0.139         | 0.096   |
| External H3K4me3         | 6-mer logistic regression | 0.586         | 0.536           | 0.103     | 0.299       | 0.371         | 0.116   |
| External H3K4me3         | CNN                       | 0.606         | 0.489           | 0.200     | 0.032       | 0.126         | 0.097   |
| External H3K4me3         | RC-aug CNN                | 0.601         | 0.510           | 0.182     | 0.010       | 0.082         | 0.061   |
| External TF (human_tf_0) | Caduceus-Ph               | 0.758         | 0.721           | 0.150     | 0.155       | 0.244         | 0.170   |
| External TF (human_tf_0) | DNABERT-2                 | 0.846         | 0.761           | 0.175     | 0.061       | 0.147         | 0.031   |
| External TF (human_tf_0) | 6-mer logistic regression | 0.782         | 0.742           | 0.117     | 0.151       | 0.217         | 0.093   |
| External TF (human_tf_0) | CNN                       | 0.703         | 0.680           | 0.099     | 0.141       | 0.229         | 0.130   |
| External TF (human_tf_0) | RC-aug CNN                | 0.823         | 0.753           | 0.159     | 0.065       | 0.132         | 0.068   |
| External TF (human_tf_1) | Caduceus-Ph               | 0.802         | 0.817           | 0.105     | 0.188       | 0.299         | 0.141   |
| External TF (human_tf_1) | DNABERT-2                 | 0.923         | 0.877           | 0.070     | 0.119       | 0.189         | 0.058   |
| External TF (human_tf_1) | 6-mer logistic regression | 0.863         | 0.831           | 0.071     | 0.097       | 0.165         | 0.084   |
| External TF (human_tf_1) | CNN                       | 0.727         | 0.711           | 0.102     | 0.132       | 0.180         | 0.117   |
| External TF (human_tf_1) | RC-aug CNN                | 0.890         | 0.868           | 0.076     | 0.125       | 0.159         | 0.045   |

Table 15: External GC-bin metrics by task, model and composition bin.

| External task    | Model                     | GC bin | GC mean | n  | P:N   | AUROC | ECE   | Brier |
|------------------|---------------------------|--------|---------|----|-------|-------|-------|-------|
| External H3K14ac | Caduceus-Ph               | 0      | 0.333   | 80 | 35:45 | 0.430 | 0.020 | 0.248 |
| External H3K14ac | Caduceus-Ph               | 1      | 0.362   | 84 | 35:49 | 0.587 | 0.076 | 0.246 |
| External H3K14ac | Caduceus-Ph               | 2      | 0.381   | 76 | 36:40 | 0.483 | 0.024 | 0.251 |
| External H3K14ac | Caduceus-Ph               | 3      | 0.401   | 84 | 43:41 | 0.545 | 0.006 | 0.249 |
| External H3K14ac | Caduceus-Ph               | 4      | 0.444   | 76 | 51:25 | 0.569 | 0.125 | 0.229 |
| External H3K14ac | DNABERT-2                 | 0      | 0.333   | 80 | 35:45 | 0.617 | 0.083 | 0.234 |
| External H3K14ac | DNABERT-2                 | 1      | 0.362   | 84 | 35:49 | 0.755 | 0.117 | 0.206 |
| External H3K14ac | DNABERT-2                 | 2      | 0.381   | 76 | 36:40 | 0.742 | 0.075 | 0.206 |
| External H3K14ac | DNABERT-2                 | 3      | 0.401   | 84 | 43:41 | 0.780 | 0.112 | 0.190 |
| External H3K14ac | DNABERT-2                 | 4      | 0.444   | 76 | 51:25 | 0.831 | 0.137 | 0.171 |
| External H3K14ac | 6-mer logistic regression | 0      | 0.333   | 80 | 35:45 | 0.571 | 0.349 | 0.358 |
| External H3K14ac | 6-mer logistic regression | 1      | 0.362   | 84 | 35:49 | 0.704 | 0.287 | 0.287 |
| External H3K14ac | 6-mer logistic regression | 2      | 0.381   | 76 | 36:40 | 0.631 | 0.271 | 0.317 |
| External H3K14ac | 6-mer logistic regression | 3      | 0.401   | 84 | 43:41 | 0.689 | 0.204 | 0.267 |
| External H3K14ac | 6-mer logistic regression | 4      | 0.444   | 76 | 51:25 | 0.715 | 0.180 | 0.240 |
| External H3K14ac | CNN                       | 0      | 0.333   | 80 | 35:45 | 0.745 | 0.092 | 0.251 |
| External H3K14ac | CNN                       | 1      | 0.362   | 84 | 35:49 | 0.623 | 0.113 | 0.254 |
| External H3K14ac | CNN                       | 2      | 0.381   | 76 | 36:40 | 0.701 | 0.056 | 0.249 |
| External H3K14ac | CNN                       | 3      | 0.401   | 84 | 43:41 | 0.648 | 0.018 | 0.248 |
| External H3K14ac | CNN                       | 4      | 0.444   | 76 | 51:25 | 0.714 | 0.140 | 0.237 |
| External H3K14ac | RC-aug CNN                | 0      | 0.333   | 80 | 35:45 | 0.722 | 0.101 | 0.242 |
| External H3K14ac | RC-aug CNN                | 1      | 0.362   | 84 | 35:49 | 0.624 | 0.118 | 0.247 |
| External H3K14ac | RC-aug CNN                | 2      | 0.381   | 76 | 36:40 | 0.655 | 0.097 | 0.243 |
| External H3K14ac | RC-aug CNN                | 3      | 0.401   | 84 | 43:41 | 0.651 | 0.083 | 0.244 |
| External H3K14ac | RC-aug CNN                | 4      | 0.444   | 76 | 51:25 | 0.624 | 0.158 | 0.239 |
| External H3K4me3 | Caduceus-Ph               | 0      | 0.331   | 86 | 37:49 | 0.653 | 0.039 | 0.238 |
| External H3K4me3 | Caduceus-Ph               | 1      | 0.357   | 76 | 35:41 | 0.608 | 0.039 | 0.246 |
| External H3K4me3 | Caduceus-Ph               | 2      | 0.377   | 86 | 42:44 | 0.505 | 0.007 | 0.251 |

## 6 Real-task Motif Probes

Table 16: Controlled real-task motif probes with GC-preserving and random-edit controls.

| Task                   | Model                        | n    | AUROC | Motif drop | GC-preserving | Random edit | Motif-random | Mono drop | Dinuc drop |
|------------------------|------------------------------|------|-------|------------|---------------|-------------|--------------|-----------|------------|
| Promoters              | 6-mer logistic               | 1000 | 0.916 | 0.015      | 0.011         | -0.005      | 0.019        | -0.003    | 0.058      |
| Promoters              | CNN                          | 1000 | 0.893 | 0.010      | 0.006         | -0.000      | 0.010        | -0.002    | 0.046      |
| Promoters              | RC-aug CNN                   | 1000 | 0.902 | 0.011      | 0.007         | 0.000       | 0.010        | -0.035    | 0.038      |
| Promoters              | DNABERT-2                    | 1000 | 0.913 | 0.006      | 0.004         | 0.001       | 0.005        | -0.054    | 0.004      |
| Promoters              | Caduceus-Ph                  | 1000 | 0.853 | 0.005      | 0.007         | -0.000      | 0.006        | -0.060    | 0.002      |
| Enhancers<br>(Cohn)    | 6-mer logistic<br>regression | 1000 | 0.720 | 0.009      | 0.012         | 0.003       | 0.006        | -0.080    | 0.138      |
| Enhancers<br>(Cohn)    | CNN                          | 1000 | 0.752 | -0.002     | 0.001         | 0.000       | -0.002       | 0.008     | 0.014      |
| Enhancers<br>(Cohn)    | RC-aug CNN                   | 1000 | 0.766 | -0.001     | 0.001         | -0.000      | -0.001       | 0.003     | 0.017      |
| Enhancers<br>(Cohn)    | DNABERT-2                    | 1000 | 0.833 | -0.000     | -0.002        | -0.001      | 0.001        | -0.049    | 0.023      |
| Enhancers<br>(Cohn)    | Caduceus-Ph                  | 1000 | 0.774 | -0.000     | 0.002         | -0.000      | 0.000        | -0.103    | -0.021     |
| Enhancers<br>(Ensembl) | 6-mer logistic<br>regression | 1000 | 0.778 | 0.034      | 0.046         | 0.004       | 0.030        | 0.203     | 0.216      |
| Enhancers<br>(Ensembl) | CNN                          | 1000 | 0.881 | 0.006      | 0.010         | 0.005       | 0.001        | 0.178     | 0.065      |
| Enhancers<br>(Ensembl) | RC-aug CNN                   | 1000 | 0.894 | 0.004      | 0.009         | 0.005       | -0.001       | 0.320     | 0.127      |
| Enhancers<br>(Ensembl) | DNABERT-2                    | 1000 | 0.807 | 0.012      | 0.019         | 0.002       | 0.010        | 0.043     | 0.009      |
| Enhancers<br>(Ensembl) | Caduceus-Ph                  | 1000 | 0.720 | -0.000     | 0.003         | -0.000      | 0.000        | -0.032    | -0.009     |

## 7 GenomeCF-Synth

Table 17: Original planted-motif synthetic benchmark summary.

| Condition     | Model          | Seeds | AUROC | ECE   | Brier | Mono drop | Dinuc drop | Motif drop |
|---------------|----------------|-------|-------|-------|-------|-----------|------------|------------|
|               | 6-mer logistic |       |       |       |       |           |            |            |
| GC correlated | regression     | 1     | 1.000 | 0.000 | 0.000 | 0.000     | 0.000      | 0.001      |
| GC correlated | CNN            | 5     | 1.000 | 0.000 | 0.000 | 0.000     | -0.000     | 0.000      |
| GC correlated | RC-aug CNN     | 5     | 1.000 | 0.000 | 0.000 | 0.000     | -0.000     | 0.000      |
|               | 6-mer logistic |       |       |       |       |           |            |            |
| GC matched    | regression     | 1     | 1.000 | 0.004 | 0.000 | 0.949     | 0.935      | 0.948      |
| GC matched    | CNN            | 5     | 1.000 | 0.040 | 0.006 | 0.876     | 0.867      | 0.870      |
| GC matched    | RC-aug CNN     | 5     | 1.000 | 0.004 | 0.000 | 0.947     | 0.938      | 0.948      |

Table 18: Extended GenomeCF-Synth shortcut-conflict and motif-grammar results.

| Task                       | Model                     | AUROC | ECE   | Brier | Mono drop | Dinuc drop | Motif drop | Rule rate | Shortcut rate |
|----------------------------|---------------------------|-------|-------|-------|-----------|------------|------------|-----------|---------------|
| GC shortcut conflict       | 6-mer logistic regression | 0.000 | 0.999 | 0.998 | 0.001     | 0.001      | 0.001      | 0.000     | 1.000         |
| GC shortcut conflict       | CNN                       | 0.000 | 0.992 | 0.991 | 0.001     | 0.002      | 0.001      | 0.008     | 0.992         |
| GC shortcut conflict       | RC-aug CNN                | 0.000 | 0.968 | 0.966 | 0.000     | 0.000      | 0.000      | 0.029     | 0.971         |
| GC shortcut conflict       | DNABERT-2                 | 0.000 | 0.994 | 0.989 | 0.000     | 0.000      | 0.001      | 0.000     | 1.000         |
| GC shortcut conflict       | Caduceus-Ph               | 0.000 | 0.995 | 0.990 | -0.000    | -0.000     | 0.000      | 0.000     | 1.000         |
| Two-motif grammar          | 6-mer logistic regression | 0.585 | 0.358 | 0.379 | 0.079     | 0.069      | -0.013     | 0.564     | 0.501         |
| Two-motif grammar          | CNN                       | 0.997 | 0.007 | 0.009 | 0.929     | 0.933      | 0.225      | 0.989     | 0.617         |
| Two-motif grammar          | RC-aug CNN                | 0.995 | 0.012 | 0.010 | 0.788     | 0.784      | 0.161      | 0.988     | 0.618         |
| Two-motif grammar          | DNABERT-2                 | 0.602 | 0.016 | 0.243 | 0.049     | 0.048      | 0.018      | 0.574     | 0.549         |
| Two-motif grammar          | Caduceus-Ph               | 0.548 | 0.014 | 0.249 | 0.001     | 0.002      | 0.002      | 0.539     | 0.526         |
| Motif identity vs position | 6-mer logistic regression | 0.981 | 0.028 | 0.029 | 0.896     | 0.885      | 0.818      | 0.970     | 0.030         |
| Motif identity vs position | CNN                       | 0.534 | 0.481 | 0.485 | -0.006    | -0.009     | -0.002     | 0.500     | 0.499         |
| Motif identity vs position | RC-aug CNN                | 0.443 | 0.495 | 0.495 | 0.005     | 0.001      | 0.003      | 0.502     | 0.497         |
| Motif identity vs position | DNABERT-2                 | 0.854 | 0.061 | 0.160 | 0.240     | 0.194      | 0.177      | 0.774     | 0.226         |
| Motif identity vs position | Caduceus-Ph               | 0.602 | 0.007 | 0.242 | 0.005     | -0.002     | 0.002      | 0.570     | 0.430         |
| GC correlated              | 6-mer logistic regression | 1.000 | 0.000 | 0.000 | 0.000     | 0.000      | 0.000      | 1.000     | 1.000         |
| GC matched                 | 6-mer logistic regression | 1.000 | 0.005 | 0.000 | 0.946     | 0.927      | 0.675      | 1.000     | 0.500         |
| GC correlated              | CNN                       | 1.000 | 0.000 | 0.000 | 0.000     | 0.000      | 0.000      | 1.000     | 1.000         |
| GC matched                 | CNN                       | 1.000 | 0.095 | 0.053 | 0.772     | 0.731      | 0.514      | 0.920     | 0.420         |
| GC correlated              | RC-aug CNN                | 1.000 | 0.000 | 0.000 | 0.000     | 0.000      | 0.000      | 1.000     | 1.000         |
| GC matched                 | RC-aug CNN                | 1.000 | 0.003 | 0.001 | 0.951     | 0.933      | 0.675      | 1.000     | 0.499         |
| GC correlated              | DNABERT-2                 | 1.000 | 0.004 | 0.000 | 0.001     | 0.001      | 0.001      | 1.000     | 1.000         |
| GC matched                 | DNABERT-2                 | 0.848 | 0.086 | 0.166 | 0.254     | 0.207      | 0.168      | 0.772     | 0.537         |
| GC correlated              | Caduceus-Ph               | 1.000 | 0.004 | 0.000 | -0.000    | -0.000     | 0.000      | 1.000     | 1.000         |
| GC matched                 | Caduceus-Ph               | 0.549 | 0.007 | 0.248 | 0.003     | -0.001     | 0.003      | 0.531     | 0.500         |

## 8 Shortcut Score and Confounder Summaries

Table 19: GenomeCF Shortcut Score summary.

| Model                     | Mean AUROC | Mean AUROC rank | Mean Shortcut Score |
|---------------------------|------------|-----------------|---------------------|
| DNABERT-2                 | 0.831      | 4.000           | 3.125               |
| RC-aug CNN                | 0.837      | 3.667           | 2.333               |
| 6-mer logistic regression | 0.778      | 2.667           | 3.167               |
| Caduceus-Ph               | 0.781      | 2.333           | 3.667               |
| CNN                       | 0.812      | 2.333           | 2.708               |

Table 20: Per-task Shortcut Score components.

| Task                | Model                     | AUROC | Shortcut score | GC-bin gap | Matched drop |
|---------------------|---------------------------|-------|----------------|------------|--------------|
| Enhancers (Cohn)    | Caduceus-Ph               | 0.773 | 4.000          | 0.082      | 0.079        |
| Enhancers (Cohn)    | DNABERT-2                 | 0.830 | 3.000          | 0.055      | 0.049        |
| Enhancers (Cohn)    | 6-mer logistic regression | 0.700 | 3.250          | 0.047      | 0.035        |
| Enhancers (Cohn)    | CNN                       | 0.754 | 2.750          | 0.068      | 0.077        |
| Enhancers (Cohn)    | RC-aug CNN                | 0.765 | 2.000          | 0.023      | 0.060        |
| Enhancers (Ensembl) | Caduceus-Ph               | 0.718 | 3.750          | 0.064      | 0.066        |
| Enhancers (Ensembl) | DNABERT-2                 | 0.775 | 3.000          | 0.027      | 0.001        |
| Enhancers (Ensembl) | 6-mer logistic regression | 0.736 | 3.500          | 0.053      | 0.019        |
| Enhancers (Ensembl) | CNN                       | 0.839 | 2.625          | 0.072      | 0.017        |
| Enhancers (Ensembl) | RC-aug CNN                | 0.884 | 2.125          | 0.062      | 0.017        |
| Promoters           | Caduceus-Ph               | 0.853 | 3.250          | 0.208      | 0.060        |
| Promoters           | DNABERT-2                 | 0.888 | 3.375          | 0.262      | 0.016        |
| Promoters           | 6-mer logistic regression | 0.898 | 2.750          | 0.218      | 0.004        |
| Promoters           | CNN                       | 0.843 | 2.750          | 0.307      | 0.018        |
| Promoters           | RC-aug CNN                | 0.864 | 2.875          | 0.335      | 0.027        |

## 9 Statistical Support and Traceability

## 10 Availability and Registry Traceability

Table 21: Confounder summary on the sampled official test subsets used for the completed benchmark budget.

| Task                | GC (neg/pos)  | CpG O/E       | Length          | Repeat          | N frac.         |
|---------------------|---------------|---------------|-----------------|-----------------|-----------------|
| Promoters           | 0.484 / 0.627 | 0.265 / 0.682 | 251.0 / 251.0   | / 0.222 / 0.243 | / 0.000 / 0.000 |
| Enhancers (Cohn)    | 0.394 / 0.469 | 0.182 / 0.258 | 500.0 / 500.0   | / 0.236 / 0.232 | / 0.000 / 0.000 |
| Enhancers (Ensembl) | 0.409 / 0.460 | 0.209 / 0.249 | 267.9 / 267.9   | / 0.233 / 0.245 | / 0.001 / 0.000 |
| Open chromatin      | 0.409 / 0.424 | 0.209 / 0.202 | 326.9 / 326.9   | / 0.234 / 0.216 | / 0.001 / 0.000 |
| Mouse enhancers     | 0.210 / 0.388 | 0.337 / 0.304 | 2500.7 / 2500.7 | / 0.591 / 0.358 | / 0.480 / 0.179 |
| Fly enhancers       | 0.420 / 0.416 | 0.926 / 0.925 | 2123.7 / 2123.7 | / 0.223 / 0.244 | / 0.000 / 0.000 |

Table 22: Human-task chromosome-fold counts on the sampled official test subsets.

| Task                | Fold A | Fold B | Fold C | Fold D | Fold E |
|---------------------|--------|--------|--------|--------|--------|
| Promoters           | 819    | 1,142  | 1,083  | 1,468  | 4,522  |
| Enhancers (Cohn)    | 517    | 926    | 878    | 1,169  | 3,458  |
| Enhancers (Ensembl) | 2,718  | 3,844  | 3,848  | 5,058  | 15,502 |
| Open chromatin      | 3,040  | 4,382  | 4,334  | 5,680  | 17,516 |

Table 23: Statistical support for the main manuscript claims. Bootstrap confidence intervals are reported where available from aggregated analyses; direct registry point estimates are retained for single-assay case-study and synthetic-mechanism claims.

| Claim                                            | Metric                      | Estimate | 95% CI         | Test                                   | $n$  | Source           |
|--------------------------------------------------|-----------------------------|----------|----------------|----------------------------------------|------|------------------|
| abstract promoter kmer auroc                     | AUROC                       | 0.898    | –              | registry point estimate                | 2000 | main results     |
| abstract promoter kmer rc instability            | RC instability              | 0.263    | –              | registry point estimate                | 2000 | main results     |
| abstract promoter rcaug rc instability           | RC instability              | 0.017    | –              | registry point estimate                | 2000 | main results     |
| cohn dnabert negative mono drop                  | Mononucleotide shuffle drop | -0.049   | –              | registry point estimate                | 6948 | main results     |
| promoter caduceus vs dnabert rc delta gain       | RC instability improvement  | 0.037    | –              | difference of registry point estimates | 2000 | main results     |
| promoter gc only matched negative drop           | Matched-negative AUROC drop | 0.084    | –              | difference of registry point estimates | 2000 | matched negative |
| promoter rcaug temperature ece reduction         | ECE reduction               | 0.031    | –              | difference of registry point estimates | 2000 | mitigation       |
| external auroc only r2                           | $R^2$                       | 0.062    | [0.001, 0.215] | bootstrap confidence interval          | 82   | external stats   |
| external shortcut only r2                        | $R^2$                       | 0.023    | [0.000, 0.111] | bootstrap confidence interval          | 82   | external stats   |
| external full profile r2                         | $R^2$                       | 0.383    | [0.263, 0.564] | bootstrap confidence interval          | 82   | external stats   |
| external full profile advantage                  | Delta $R^2$ over AUROC-only | 0.321    | [0.182, 0.515] | bootstrap confidence interval          | 82   | external stats   |
| case a bcl11a auprc gain                         | AUPRC gain                  | 0.069    | –              | difference of registry point estimates | 1    | case study       |
| case a bcl11a topk gain                          | Top-k enrichment gain       | 0.823    | –              | difference of registry point estimates | 1    | case study       |
| case b myc dnabert topk advantage                | Top-k enrichment advantage  | 0.430    | –              | difference of registry point estimates | 1    | case study       |
| synthetic gc conflict dnabert shortcut following | Shortcut-following rate     | 1.000    | –              | deterministic synthetic evaluation     | 1    | synthetic        |

Table 24: Key release artifacts and local paths.

| Artifact                         | Local path                                            |
|----------------------------------|-------------------------------------------------------|
| Canonical registry               | results/release/benchmark_registry.csv                |
| Release summary                  | results/release/benchmark_summary.csv                 |
| Publication tables               | results/publication/*.csv                             |
| Publication TeX tables           | results/publication/*.tex                             |
| Main manuscript PDF              | paper/genomecf_report.pdf                             |
| Supplementary PDF                | paper/genomecf_supplement.pdf                         |
| Release validation report        | results/release/validation_report.json                |
| Statistical claims table         | results/release/statistical_claims.csv                |
| Paper claims YAML                | paper/claims.yaml                                     |
| Release bundle manifest          | release/GenomeCF_v1_manifest.json                     |
| Release bundle checksums         | release/GenomeCF_v1_checksums.txt                     |
| Nature Methods artifact manifest | results/release/nature_methods_artifact_manifest.json |